PARTICIPANT INFORMATION SHEET & CONSENT FORM

(For Social, Behavioural and Educational Research studies)

1. Protocol title

A route-choice questionnaire on walking in hot hours in Singapore (Walking in the shade)

2. Principal Investigator and co-investigator(s), if any, with the contact number and organization:

PI: Yu Qian Ang
Contact number: (65) 6516 1008 · Email: yuqian@nus.edu.sg
Organization: Department of the Built Environment, College of Design and Engineering, National University of Singapore

Co-PI: Shuyang Zhang
Email: shuyang.zhang@nus.edu.sg
Organization: Department of the Built Environment, College of Design and Engineering, National University of Singapore

3. What is the purpose of this research?

You are invited to participate in a research study. This information sheet provides you with information about the research study. The Principal Investigator (the person in charge of this research) or his/her representative will also describe this research to you and answer all of your questions. Read the information below and ask questions about anything you don't understand before deciding whether or not to take part.

Walking in Singapore's hot weather is more comfortable in the shade. Besides trees, the city provides shade facilities such as arcades (five-foot ways), covered linkways, covered overhead bridges and corridors between buildings. This study aims to help improve these shade facilities. We want to understand how people choose their walking route on hot days: for example, whether they would walk a little further to stay in the shade, and how much further. Your answers will help us check a computer model that suggests shadier walking routes.

4. Who can participate in the research? What is the expected duration of my participation? What is the duration of this research?

You can take part if you are a member of the public, an NUS student or an NUS staff member, and you:

- are aged 21 to 64;
- have lived in Singapore for at least 6 months; and
- can read English or Simplified Chinese and complete a short questionnaire on your own phone.

You cannot take part if you:

- are under 21 or aged 65 or above;
- have lived in Singapore for less than 6 months;
- are currently taught, tutored, supervised or graded by a member of the research team, or report to a member of the research team;
- are a member of the research team; or
- have already completed this questionnaire.

Your participation is a single session of about 5 minutes, on the spot. There are no follow-up sessions. Data collection will take place on sunny weekdays between 11:00 and 15:00, within about one week after ethics approval.

5. What is the approximate number of research participants involved?

About 150 participants will take part in this research.

6. What will be done if I take part in this research study?

A research assistant will ask whether you are aged 21 to 64 and have lived in Singapore for at least 6 months. If you agree to take part, you will scan a QR code and answer an anonymous questionnaire on your own phone, in English or Simplified Chinese, in a shaded spot nearby. The questionnaire has 8 questions:

- your gender, your age band and whether you are an NUS student, an NUS staff member or neither, and how you use and notice shade facilities (with photos);
- for one real walking trip, which of two routes you would take (a shorter route or a shadier route), first from short descriptions and then from information shown as in a navigation app;
- for three real walking trips, which of seven routes you would prefer, from the shortest to the shadiest.

All questions are about choices on the screen; you will not be asked to walk any route. You can go back to change your earlier answers before you submit. After you submit, you will see a completion screen with an 8-character code; please show it to the research assistant to receive your payment.

7. How will my privacy and the confidentiality of my research records be protected?

No personal data is collected in the survey. We do not ask for your name, contact details, address, NRIC number or location, and we do not record your device information or the exact time you answer. The questionnaire only asks for your gender, a broad age band (for example, 25–34) and whether you are an NUS student, an NUS staff member or neither. Your answers are stored under a random code that cannot be linked to you: first in a Google Sheet that only the research team can access, and then on NUS storage after data collection ends. Results will be reported only as group totals.

All data collected will be kept in accordance to the University's Research Data Management Policy. Research data used in any publication will be kept for a minimum of 10 years before being discarded.

8. What are the possible discomforts and risks for participants?

No significant discomfort and risks are expected for participants as their participation lies only in the completion of the surveys. As the survey takes place outdoors around midday, the research assistant will invite you to answer it in a shaded spot, and you may stop at any time.

9. What is the compensation for any injury?

As no significant discomfort and risks are expected for participants, there is no compensation provided.

If you follow the directions of the PI in charge of this research study and you are injured, the NUS will pay the medical expenses for the treatment of that injury. By giving your consent, you will not waive any of your legal rights or release the parties involved in this study from liability for negligence.

10. Will there be reimbursement for participation?

Yes. You will receive SGD 5 in cash on the spot after you complete and submit the questionnaire and show the completion screen to the research assistant. There is only one session, so the payment is not pro-rated. If you stop before submitting, no payment is made, and this does not affect you in any other way. The research assistant only looks at the 8-character code on your screen to confirm that you have finished; the code is not written down or recorded, and we do not record your name, contact details or signature.

11. What are the possible benefits to me and to others?

There is no direct benefit to you by participating in this research study. The knowledge gained may benefit the public in the future: it may help planners decide where shade facilities such as covered linkways and arcades are most needed, and help navigation tools suggest shadier walking routes on hot days.

12. Can I refuse to participate in this research?

Yes, you can. Your decision to participate in this research study is voluntary and completely up to you. You can also withdraw from the research at any time before you submit your answers, without giving any reasons, simply by closing the page; your unfinished answers will not be submitted and no data will be kept. Because the survey is anonymous, once you have clicked “Submit” it is not possible to withdraw your answers, as they cannot be linked back to you.

13. Whom should I call if I have any questions or problems?

Please contact the Principal Investigator, Yu Qian Ang (telephone (65) 6516 1008, email yuqian@nus.edu.sg), or Attn: Shuyang Zhang (email shuyang.zhang@nus.edu.sg) for all research-related matters and in the event of research-related injuries.

For an independent opinion specifically regarding the rights and welfare of research participants, you may contact a staff member of the College of Design and Engineering Ethics Review Committee (CDE ERC) at cdebox5@nus.edu.sg.

Online Consent Form

Protocol title:
A route-choice questionnaire on walking in hot hours in Singapore (Walking in the shade)

Principal Investigator with the contact number and organization:
Yu Qian Ang, Department of the Built Environment, College of Design and Engineering, National University of Singapore. Telephone (65) 6516 1008, email yuqian@nus.edu.sg.

I hereby acknowledge that:

1. I have agreed to take part in the above research.
2. I have received a copy of this information sheet that explains the use of my data in this research. I understand its contents and agree to donate my data for the use of this research.
3. I am aged 21 to 64 and have lived in Singapore for at least 6 months.
4. I can withdraw from the research at any point of time before submitting, by closing the page, and my unfinished answers will not be kept. However, for online anonymous surveys, I understand that once I have clicked “Submit”, it is not possible to withdraw as responses cannot be linked back to me.
5. I will receive SGD 5 in cash only after I complete and submit the questionnaire. If I do not agree, or stop before submitting, I will not receive the payment.
6. I will not have any financial benefits that result from the commercial development of this research.

☐ I agree to the information provided in the Participation Information Sheet and consent to participate.

☐ Disagree. Exit survey.